Van Kieu Dang (Kiel)

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EDUCATION

Northeastern University MA, USA
Master – Analytics, Applied Machine Intelligence, College of Professional Studies Sep 2021 – Jul 2023
GPA: 4.0/4.0 (97.89%) - Valedictorian.
Capstone Project: Computer vision: Unsafe excavation detection for construction safety - Improving construction site safety through the integration of multi-sensor fusion and deep learning-based object detection.
Advisor: Dr. Sohom Mandal, Faculty of Analytics, College of Professional Studies, Northeastern University.

Foreign Trade University Ho Chi Minh City, Vietnam
Bachelor – International Business Administration, Faculty of Business Administration Sep 2011 – Feb 2015
GPA: 3.34/4.0 (8.28/10) – Top 5%

RESEARCH INTERESTS

My research passion centers on advancing machine learning's impact in the industry, encompassing Deep Learning, Computer Vision, Natural Language Processing (NLP), Time Series, Causal Frameworks, Trustworthiness and Privacy. Additionally, I'm intrigued by the intersection of those domains – an interdisciplinary approach, seeking novel solutions to complex problems that transcend traditional boundaries.

PROFESSIONAL EXPERIENCE

Hitachi Vantara Corporation CA, USA
Senior Data Scientist (Remote Full time) Oct 2023 – Present

- Pioneered the application of Deep Learning methodologies to deploy and operationalize advanced predictive analytics models, catering to diverse industries such as healthcare, retail, manufacturing, and finance.
- Spearheaded the development and integration of NLP techniques, leveraging cutting-edge algorithms to enhance text analytics capabilities, particularly beneficial for clients in the finance sector.
- Developed novel time series forecasting techniques using causal frameworks, improving supply chain optimization for retailers and reducing stockouts.

A Medium Corporation CA, USA
Machine Learning Content Engineer (Freelancer – Partner Program) Nov 2022 – Present

- Translated complex machine learning topics, particularly in NLP and time series, into accessible and insightful content for a diverse audience.

Definity Financial ON, Canada
Data Scientist and Modeling (Co-op) Jan 2023 – May 2023

- Executed two comprehensive dataset preparations using image and text data related to car accidents and tabular data for predictive model enhancement.
- Developed and adapted 4 predictive models for underwriting, actuary, and claims handling, following pipeline and ensuring code reproducibility.
- Secured victory in a Hackathon by successfully implementing an NLP and MLOps project for Form Autofill using Optical Character Recognition and Language Model.

Alibaba Group – Lazada E-commerce Ho Chi Minh City, Vietnam
Senior Manager, Category Management Sep 2020 – Aug 2021

- Collaborated with the Data Science team to identify and weigh factors influencing search exposure in e-commerce platforms by employing advanced data analysis methodologies like text mining and to conduct A/B testing to evaluate the impact of key sales drivers (free shipping, vouchers, and hot deals) on revenue growth.
- Conducted ad-hoc analysis, applying hypothesis testing to answer business inquiries arising from day-to-day operations.

TEACHING EXPERIENCE

Teaching Assistant of Introduction to Statistics and Probability (Fall 2021), College of Professional Studies, Northeastern University.

HIGHLIGHTED PROJECTS

Customer Service Chatbot with In-Context Learning | *Python, Llama Index, OpenAI, Hugging Face | Github and Blog*

- This project is a deep dive into the world of AI-driven customer service chatbots, enhanced by the power of in-context learning. We leverage the Llama Index and Language Model API to create a chatbot that understands and responds to

customer inquiries effectively, transforming the way businesses provide support.

Automated Text Data Extraction and Form Filling System | *Python, NLP, OpenAI* | *Github and Blog*

- By replacing manual form filling with AI-driven automation, this application empowers our clients to obtain quotes effortlessly, fostering rapid business growth and enhancing customer experience.
- Utilized Regular Expressions, Rule-based extraction, Named Entity Recognition (NER), and LayoutLMv3 to extract essential data from unstructured documents and automate the insurance forms filling for our clients.

Network Analysis and Community Detection | *Python, SpaCy, NER, NetworkX* | *Github and Blog*

- This project explores "Dune" book series, leveraging Python, SpaCy, and NER for Named Entity Recognition. The author extracts character relationships, performs network analysis using NetworkX and the Louvain Algorithm, assesses modularity, and identifies influential characters through centrality measures, offering fresh insights into this science fiction masterpiece.

Resume and Job Matching | *Python, Doc2Vec* | *Github, and Blog*

- This application predicts the matching percentage of a candidate's resume to a job posting. It utilizes the Doc2Vec model and the Gale-Shapley algorithm.
- A nice visualization and notification tool was also launched to help candidates improve their CVs to get better chances.

Real-time Object Detection with Images and Sensors Fusion | *Python, Roboflow, NVIDIA Jetson* | *Github, and App*

- This project is designed to identify unsafe holes on construction sites, helping to ensure the well-being of workers and the integrity of job sites.
- Leveraged pre-trained models for object detection like YOLO, Mask R-CNN, and Roboflow platform through the integration of Multi-Sensor Fusion to automate the workflow of real-time object detection.
- Achieved 91% mAP, 95% Precision, and 87% Recall.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL, SPSS, and Git.
- **Databases / Big Data:** SQL Server, Hadoop, and Spark.
- **Tools and Framework:** TensorFlow, PyTorch, Keras, Transformers, Llama Index, LangChain, SpaCy, NER, Nltk.
- **Specializations:** Deep Learning, NLP, Computer Vision, Casual Framework, Time Series, and MLOps.

HONORS & AWARDS

Valedictorian of the Master in Analytics – Fall 2021 Cohort, Northeastern University. Jul 2023
Hackathon Winner – OCR and Language Model for Form Autofill, Definity Financial. Feb 2023
Top 20 Finalist – VinUniversity Global Case Competition (VGCC), VinUniversity. Dec 2021
Merit-based Scholarship, 2013 and 2014, Foreign Trade University.
Talent Incubation Scholarship (for Top 5 students of the faculty), Coca-Cola Vietnam and Thanh Nien News. Nov 2013
Ambassador – Exchange participant in Indonesia, AIESEC. Jun 2013 – Aug 2013

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Mobile Food Bank service, Red Cross Toronto, Canada Mar 2023
Sorted and classified food into portioned servings, ensuring equal access for all.

MLops World Conference & Expo 2022 Volunteer, Toronto, Canada Jun 2022
Played a crucial role as an organization support volunteer, ensuring smooth operations and efficient coordination at every step.

Course Registration and Orientation Day – Winter 2022 Volunteer, Northeastern University, Toronto, Canada Dec 2021
Coordinated with the school to help new students with course registration for the new academic semester and orientation day.

REFERENCES

Sohom Mandal
Northeastern University
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Matthew Goodwin
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